Internship Experience Report: Benchmark Execution and Tool Findings

CATEC satellite simulation evaluation

July 14, 2026

1 Benchmark concepts tested

Summary table

Benchmark concept	Tools	Key outputs
Two-body orbit propagation validation	GMAT, Basilisk	Cross-tool runners; GMAT mission script; archived state histories and validation plots
Force-model fidelity build-up (perturbation ladder)	GMAT, Basilisk, Orekit, FreeFlyer	Case ladder runners (four tools); drift metrics and cross-tool comparison plots
Mission analysis with Madrid contacts and eclipse windows	GMAT, Basilisk, FreeFlyer	Contact and eclipse analysis runners; Madrid pass tables and timeline plots
Earth-observation imaging opportunity analysis	GMAT, Basilisk, Orekit	RAAN trade study; imaging opportunity counts; off-nadir and coverage metrics
LEO-to-GEO electric-transfer trajectory design	GMAT	GMAT electric-spiral design; exported burn schedule and reference ephemeris
Realistic transfer execution (design vs execution)	Basilisk, NASA 42	Basilisk execution; NASA 42 full-mission execution; design-to-execution tables
Power-aware electric propulsion (dynamic solar and eclipse)	GMAT, Basilisk, NASA 42	Dynamic solar-power model; thrust and power time series; duty-cycle metrics
Attitude-control and thrust-pointing performance	Basilisk, NASA 42	Pointing-error histories; NASA 42 ADCS telemetry; thrust-alignment plots
Reaction-wheel and momentum-management behavior	Basilisk, NASA 42	RW speed and torque diagnostics; MTB dipole commands; actuator limit checks
Cross-tool trajectory agreement against a common reference	GMAT ref, Basilisk, NASA 42	Reference ephemeris comparison; paired propulsion and eclipse plots
Long-duration end-to-end transfer campaign (∼53 days)	GMAT, Basilisk, NASA 42	Full-mission archives; fuel and pointing summaries
Constellation divergence and replay workflows	GMAT, Basilisk, FreeFlyer	Multi-spacecraft propagation scripts; state-history exports; replay recordings

Benchmark concept overview

- **Two-body orbit propagation validation** — verified that tools reproduce analytical Kepler motion from common initial conditions with very small state drift. *Tested: GMAT, Basilisk.*
- **Force-model fidelity build-up (perturbation ladder)** — quantified orbit divergence and runtime as perturbations were added progressively (J2, drag, SRP, Sun/Moon gravity). *Tested: GMAT, Basilisk, Orekit, FreeFlyer.*
- **Mission analysis with ground contacts and eclipse windows** — computed operational pass and eclipse metrics for a **Madrid-ground-station scenario** (including revisit and pass duration behavior). *Tested: GMAT, Basilisk.*
- **Earth-observation imaging opportunity analysis** — evaluated off-nadir geometry, access opportunities, and target-area coverage for a nadir-pointing camera concept. *Tested: GMAT, Basilisk, Orekit.*
- **LEO-to-GEO electric-transfer trajectory design** — designed a long-duration low-thrust spiral transfer with event-based convergence checks and fuel accounting. *Tested: GMAT.*
- **Realistic transfer execution (design-vs-execution check)** — executed the designed transfer with higher-fidelity spacecraft effects (attitude dynamics, thrust pointing losses, environmental coupling) to check how much real execution deviates from ideal design. *Tested: Basilisk, NASA 42 (four reaction wheels, facet mesh, EGM96 18×18 gravity).*
- **Power-aware electric propulsion (dynamic solar and eclipse)** — replaced constant effective power assumptions with power that changes with Sun distance, eclipse state, and panel model coefficients, then propagated thrust and mass-flow consequences. *Tested: GMAT (native), Basilisk, NASA 42 (binary eclipse model).*
- **Attitude-control and thrust-pointing performance** — measured how pointing error and attitude behavior affect effective along-track thrust delivery during long burns. *Tested: Basilisk, NASA 42 (gyros, star tracker, closed-loop attitude control).*
- **Reaction-wheel and momentum-management behavior** — evaluated long-run attitude-control robustness including wheel dynamics and momentum-unloading strategies. *Tested: Basilisk (RW/MTB campaigns), NASA 42 (four-wheel pyramid, MTB unload).*
- **Cross-tool trajectory agreement against a common reference** — compared propagated states to a shared reference ephemeris across long-horizon transfer scenarios. *Tested: design reference vs Basilisk free propagation and NASA 42 propagated state.*
- **Long-duration end-to-end transfer campaign (∼53 days)** — executed complete mission-horizon runs and evaluated final orbit achievement, fuel consumption, duty cycle, and run-history outputs (time series such as state, thrust, mass, power, eclipse, and pointing metrics). *Tested: GMAT (design baseline), Basilisk, NASA 42.*
- **Constellation divergence and replay workflows** — implemented multi-spacecraft propagation and state-history export workflows for perturbation-divergence analysis and replay/inspection of archived trajectories. *Tested: GMAT, Basilisk, FreeFlyer.*

1.1 Two-body orbit propagation validation

Setup	24 h propagation, 600 s samples, 500 km SSO-like initial orbit, independent Kepler analytical reference
Tools	GMAT, Basilisk
Metrics	Max/final position error vs analytical reference; max velocity error; cross-tool state difference at same epochs
Result	Both passed sub-meter accuracy: GMAT 0.249 m max position error, Basilisk 0.441 m; max GMAT–Basilisk position difference 0.689 m

Both tools reproduced two-body motion with sub-meter agreement against the analytical reference over a full 24 h arc. The remaining GMAT–Basilisk gap stayed below 1 m and is consistent with

integrator/gravity representation differences, not a failed propagation pipeline.

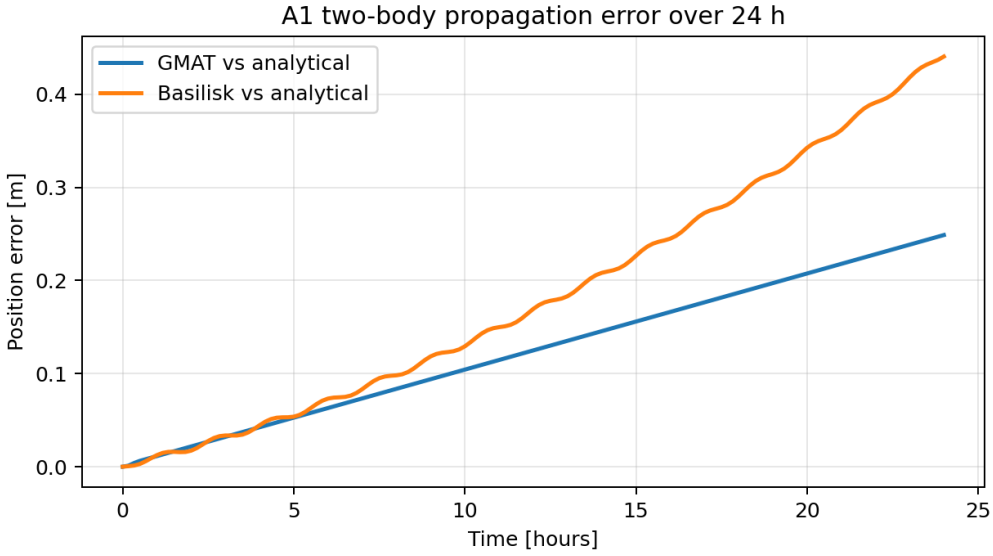


Figure 1: A1 two-body validation: position error versus analytical reference (24 h).

1.2 Force-model fidelity build-up (perturbation ladder)

Setup	A0 ladder with progressive force-model complexity (Case 1: two-body; Case 2: J2; Case 3: J2+drag; Case 4: J2+drag+SRP; Case 5: J2+drag+SRP+third-body) over long-arc propagation
Tools	GMAT, Basilisk, Orekit
Metrics	Incremental drift versus previous case, orbital-element deltas, runtime scaling with added perturbations
Result	The 24 h incremental comparison shows the dominant jump at Case 2 (J2), followed by smaller increments from drag, SRP, and third-body effects

The 24 h ladder shows J2 as the primary contributor to short-horizon divergence from two-body propagation, while drag, SRP, and third-body terms contribute smaller incremental shifts at this altitude and configuration. GMAT, Basilisk, and Orekit follow the same trend and remain closely aligned across the perturbation sequence.

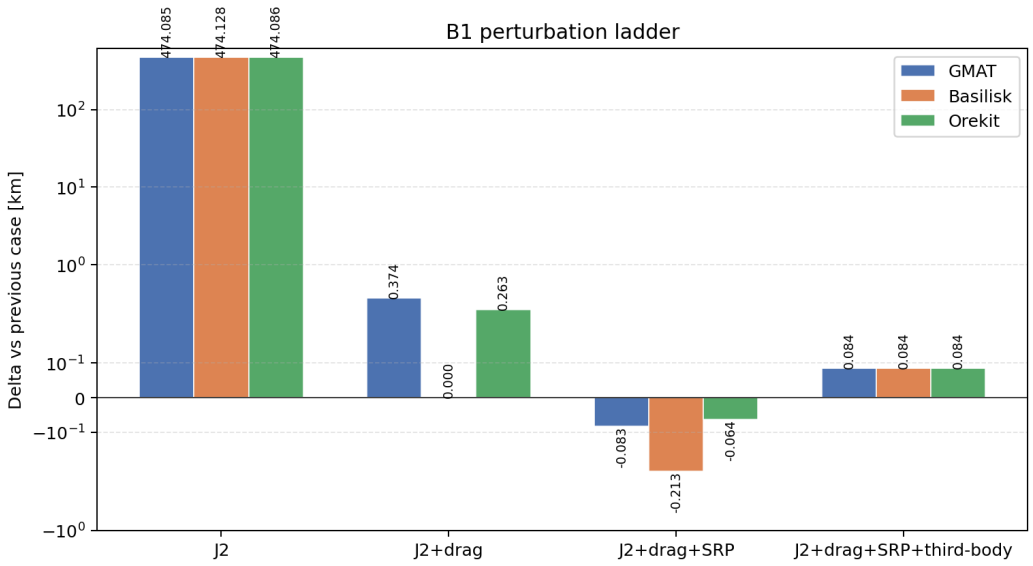


Figure 2: Incremental drift by case for GMAT, Basilisk, and Orekit (each case relative to the previous one; symlog y-axis).

1.3 Mission analysis with Madrid contacts and eclipse windows

Across tools, the Madrid contact timeline is operationally consistent: identical pass count and closely matched revisit/pass-duration behavior indicate that station-access geometry is reproduced with small implementation-level spread. The eclipse fraction agreement is also tight, so the three pipelines support comparable power-availability interpretation over the same 24 h scenario. The FreeFlyer run uses a different mission context that includes non-Madrid imaging events and a small nonzero initial eccentricity, which explains why it is treated here as consistency evidence rather than strict one-to-one parity.

Metric	GMAT	Basilisk	FreeFlyer
Madrid pass count	4	4	4
Avg pass duration [min]	5.530	5.458	5.312
Min revisit [min]	94.182	94.167	94.250
Eclipse time [%]	14.830	15.243	14.635

Table 2: A2 mission-analysis metrics for Madrid access and eclipse occupancy (24 h window).

1.4 Earth-observation imaging opportunity analysis

The RAAN sweep isolates a single operating point that satisfies the A3 imaging rule with full regional coverage and a valid access window in the tested box. At the selected RAAN, all tools converge on the same opportunity-count outcome and nearly identical off-nadir geometry, which indicates that targeting logic and camera-intersection handling are aligned well enough for cross-tool comparability.

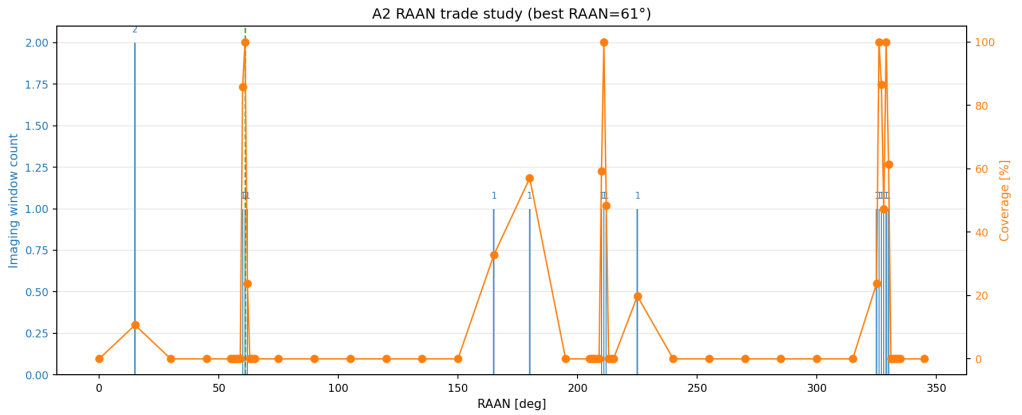


Figure 3: RAAN trade-study evidence used for selecting the RAAN = 61° A3 case.

Metric	GMAT (RAAN 61)	Basilisk (RAAN 61)	Orekit (RAAN 61)
Imaging opportunity count	1	1	1
Minimum off-nadir [deg]	2.175	2.110	2.175
Camera FOV full cone [deg]	10	10	10
Madrid regional coverage [%]	67.03	67.93	67.03
Coverage max count [samples/cell]	2	2	2

Table 3: A3 imaging metrics with common nadir-camera geometry and selected RAAN = 61°.

1.5 LEO-to-GEO electric-transfer trajectory design

The design result establishes a feasible low-thrust transfer baseline that reaches the GEO-size semi-major-axis target with bounded propellant usage over a multi-week spiral. Because the objective is SMA-only convergence, the achieved endpoint is expected to retain residual eccentricity and inclination; this is a deliberate staging decision that separates energy-raising from later orbit-shaping closure.

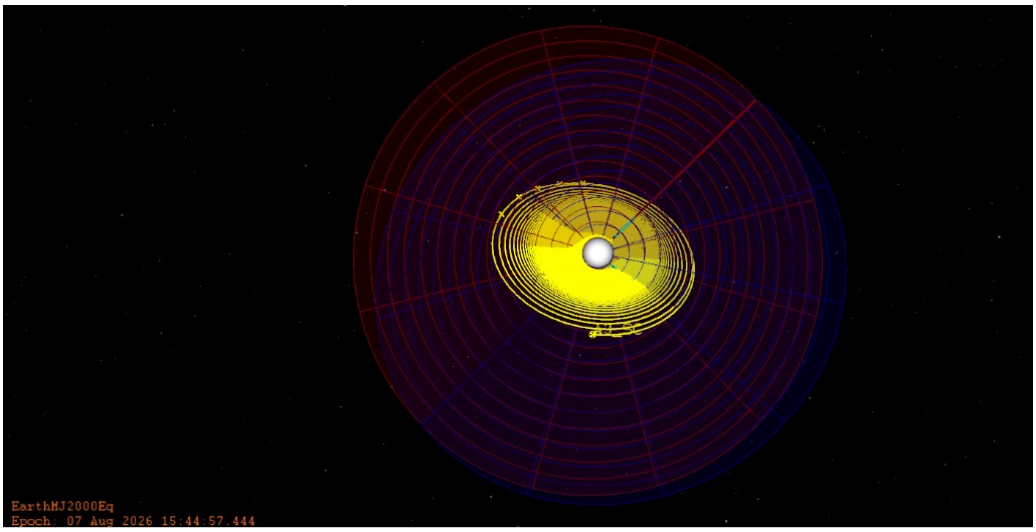


Figure 4: LEO-to-GEO electric-spiral transfer trajectory for the A3 design case.

Design metric	Value
Initial semi-major axis [km]	6878.137
Target semi-major axis [km]	42164.000
Elapsed time to SMA target [days]	52.477
Mass at start [kg]	17.000
Mass at SMA target [kg]	13.234
Propellant used to SMA target [kg]	3.766
Eccentricity at SMA target	0.0765
Inclination at SMA target [deg]	28.31

Table 4: A3 LEO-to-GEO electric-transfer design results for the SMA-targeted milestone.

1.6 Realistic transfer execution (design vs execution)

The design-to-execution assessment compares Basilisk and NASA 42 execution of the exported electric-spiral schedule. Basilisk propagates the spacecraft freely with full perturbations. NASA 42 propagates six-degree-of-freedom dynamics with a spline-guided reference orbit, four reaction wheels, facet mesh aerodynamic and solar-pressure effects, and an 18×18 Earth gravity field. Both executors match the commanded burn window within one second and remain within 0.1% of the design propellant use.

Metric	Basilisk execution	NASA 42 execution
Final SMA [km]	50227.100	41844.496
Final eccentricity [-]	0.02833	0.04857
Final inclination [deg]	28.383	28.495
Fuel used [kg]	3.9692	3.7668
Burn start timing error [s]	0.0	0.0
Max thrust-pointing error [deg]	2.311	8.724

Table 5: A3 transfer execution comparison (Basilisk vs NASA 42). Design reference at SMA target: 42 164 km SMA, 3.766 kg propellant.

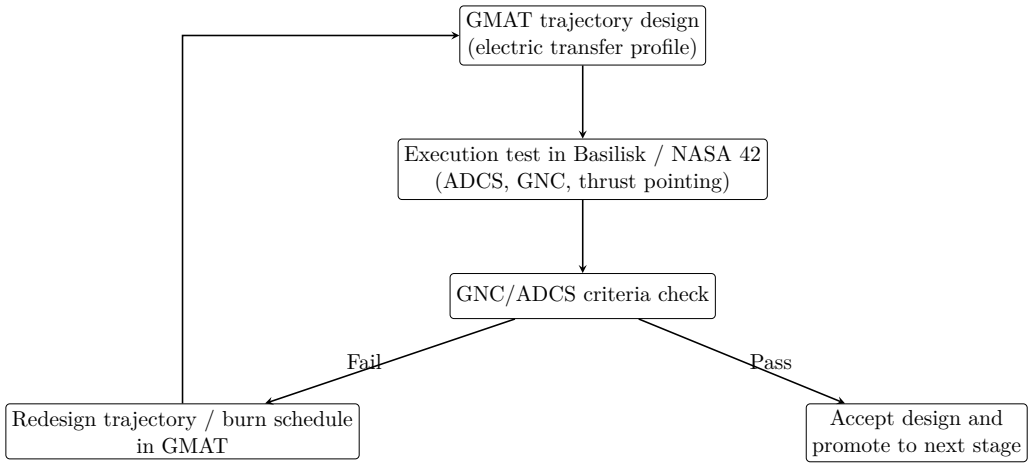


Figure 5: A3 transfer pipeline: GMAT design, high-fidelity execution test, and redesign loop on GNC/ADCS failure.

1.7 Power-aware electric propulsion (dynamic solar and eclipse)

Both execution environments apply the same solar-power and thrust–mass-flow polynomials, but they gate electrical power differently. Basilisk scales generation with a continuous Earth eclipse illumination factor (dual-cone penumbra), yielding near-unity thrust duty ($\sim$100%) and higher mean

spiral thrust and power (17.2 mN, 0.819 kW). NASA 42 retains a binary sun-on/sun-off eclipse flag for thruster solar shadowing, producing ∼87% duty, ∼13% eclipse-off time, and lower mean spiral levels (15.0 mN, 0.713 kW). Propellant expenditure is 3.969 kg in Basilisk and 3.767 kg in NASA 42. Figure 6 contrasts the paired thrust, power, and eclipse channels over the full campaign.

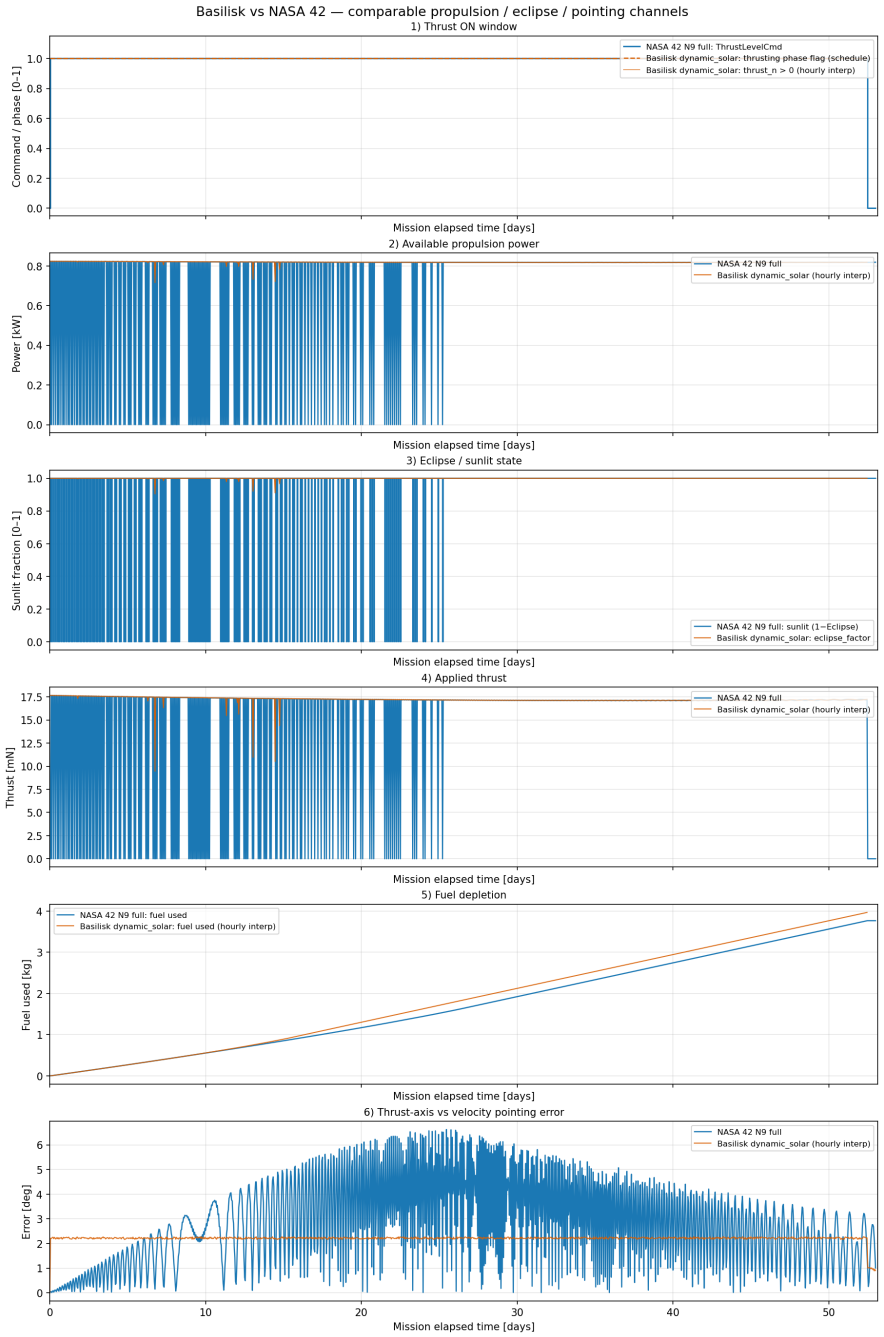


Figure 6: Basilisk vs NASA 42 power-aware electric-propulsion behavior during transfer execution: thrust, electrical power, and eclipse state over mission time.

Power-aware EP metric	Basilisk execution	NASA 42 execution
Mean spiral thrust [N]	0.01722	0.01499
Nominal thrust [N]	0.01485	0.01485
Mean spiral power [kW]	0.8190	0.7129
Nominal reference power [kW]	0.7910	0.7910
Thrust duty fraction [-]	1.0000	0.8705
Eclipse fraction [%]	0.035	12.77
Fuel used [kg]	3.9692	3.7668

Table 6: A3 power-aware electric-propulsion metrics from the ∼53-day full-mission execution (Basilisk vs NASA 42).

1.8 Attitude-control and thrust-pointing performance

Over the full ∼ 53-day campaign, both execution environments maintain velocity-oriented thrust pointing under closed-loop attitude control. Figures 7 and 8 show thrust-to-velocity alignment, reference-frame tracking error, body-rate magnitude, and attitude-error components. Basilisk holds thrust-axis misalignment near 2.2° with a 2.311° peak during spiral thrusting. NASA 42 reaches a mean thrust-axis error of 4.45° and a peak of 8.72° with four reaction wheels, three-axis gyros, star tracker, and mesh environmental torques. Nadir alignment remains bounded (mean 2.29°, max 6.05°).

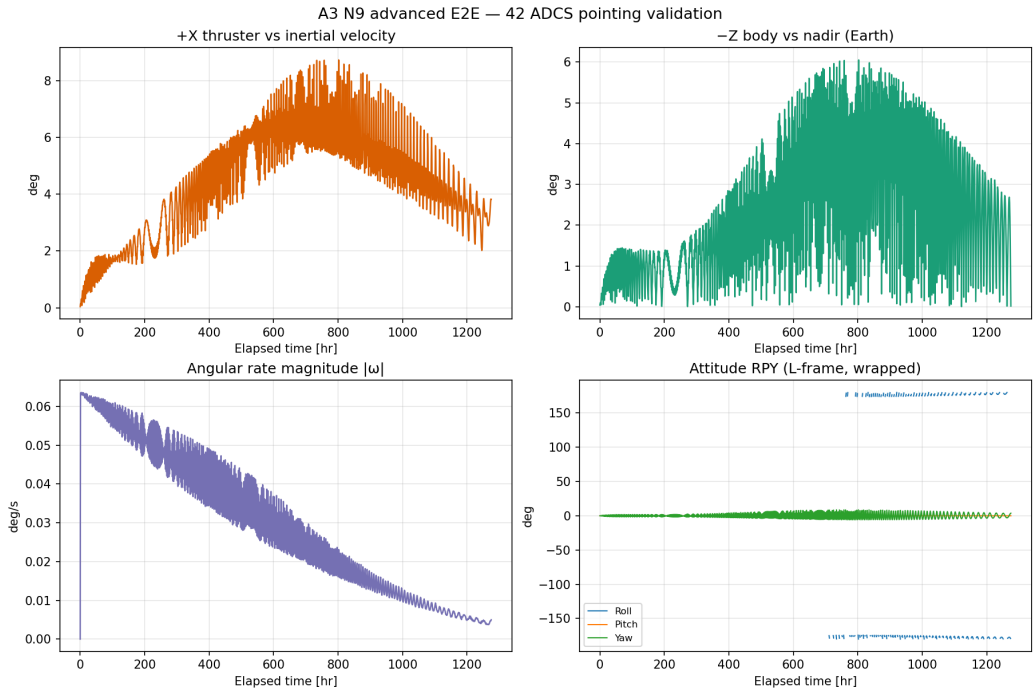


Figure 7: NASA 42 ADCS pointing during transfer execution: thrust-to-velocity alignment, nadir alignment, angular-rate magnitude, and attitude angles over mission time.

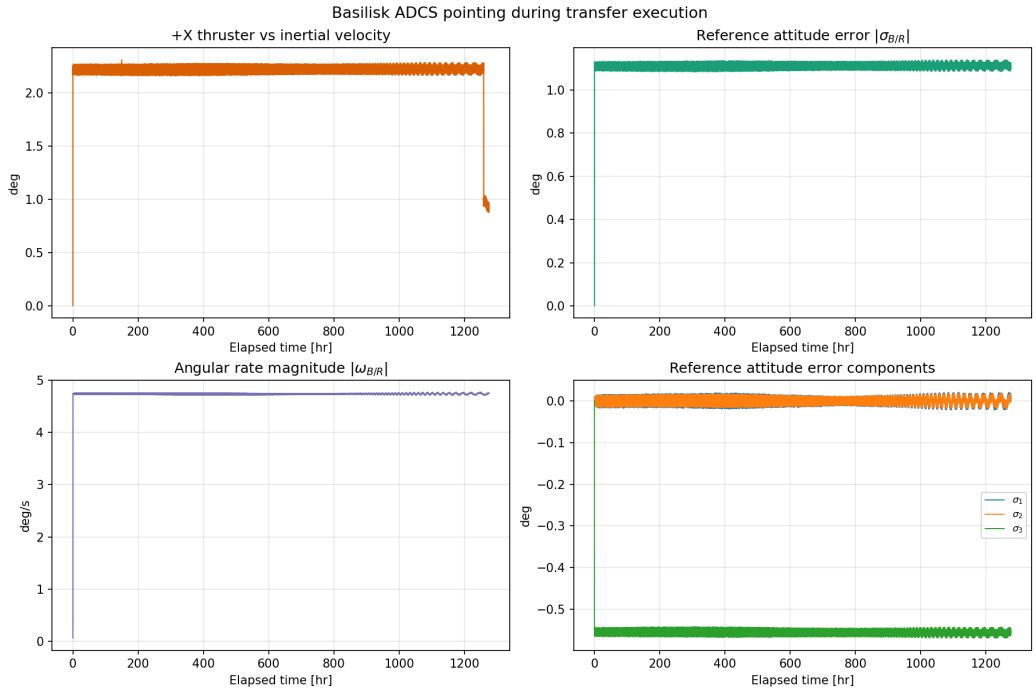


Figure 8: Basilisk ADCS pointing during transfer execution: thrust-to-velocity alignment, reference attitude error, angular-rate magnitude, and tracking-error components over mission time.

Pointing metric	Basilisk execution	NASA 42 execution
Max thrust-pointing error [deg]	2.311	8.724
Mean thrust-pointing error [deg]	2.222	4.448
P95 thrust-pointing error [deg]	2.267	5.917
Mean nadir-pointing error [deg]	—	2.285
Max nadir-pointing error [deg]	—	6.049
Peak body rate [deg/s]	—	0.064
Star-tracker valid fraction [%]	—	90.36

Table 7: A3 thrust-axis vs inertial-velocity pointing error during electric-spiral thrust intervals (Basilisk vs NASA 42).

1.9 Reaction-wheel and momentum-management behavior

Reaction-wheel and magnetorquer telemetry from the full-mission executions use the same four-wheel pyramid layout, 0.150 N m s per-wheel momentum limit, and 5.0 A m^2 per-axis MTB limit in both executors. Basilisk peaks at 7.7% of the wheel envelope with 25% RW torque saturation during the spiral and MTB unloading at the configured dipole limit. NASA 42 reaches 59% of the wheel envelope while MTB commands stay more than five orders of magnitude below that same limit.

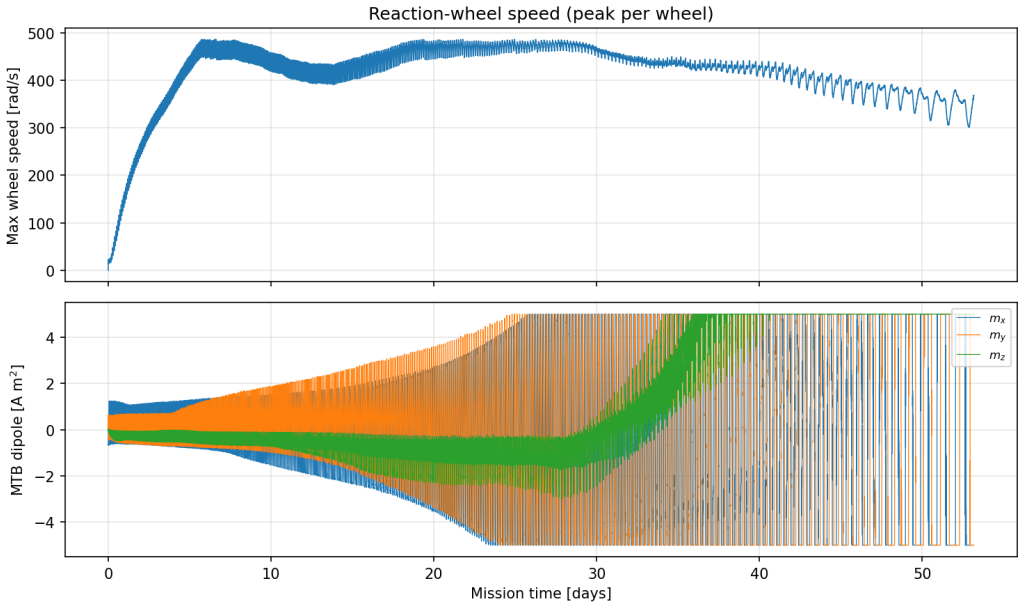


Figure 9: Basilisk reaction-wheel and magnetorquer behavior during execution: wheel-speed evolution and MTB dipole-command history.

Momentum-management metric	Basilisk execution	NASA 42 execution
RW / MTB layout	4-wheel pyramid + 3-axis MTB	4-wheel pyramid + 3-axis MTB
Wheel momentum limit [N m s]	0.150	0.150
Peak wheel momentum [N m s]	0.0116	0.089
Peak momentum / limit [%]	7.7	59.3
RW torque saturation during spiral [%]	25.0	—
MTB axis dipole limit [A m^2]	5.0	5.0
Peak MTB dipole command [A m^2]	5.00	1.1×10^{-5}
MTB dipole norm p95 [A m^2]	8.66	8.8×10^{-6}
MTB p95 norm / limit [%]	100.0	0.00010

Table 8: A3 reaction-wheel and magnetorquer comparison (Basilisk vs NASA 42). Limits are common to both configurations; MTB p95 norm is relative to the full three-axis envelope $\sqrt{3} \times 5.0$ A m^2.

The contrast follows from how each executor couples attitude control to the transfer. Basilisk propagates the spacecraft freely under thrust-pointing GNC, so environmental and thrust-misalignment torques are rejected entirely on the reaction-wheel stack; when wheel torque saturates, magnetorquer unloading runs at the dipole limit and keeps stored momentum low. NASA 42 tracks a reference-guided six-degree-of-freedom trajectory on the facet mesh, which yields a smaller disturbance torque budget and slower wheel-speed growth; wheel momentum rises toward the mid-range of the envelope but remains below the level that triggers sustained magnetorquer unloading, so dipole commands stay near zero despite identical actuator limits.

1.10 Cross-tool trajectory agreement against a common reference

Both executors follow the exported burn schedule. Basilisk raises semi-major axis through free orbit propagation. NASA 42 propagates six-degree-of-freedom spacecraft dynamics on a facet mesh with environmental torques and a spline-guided reference orbit. Figure 10 compares semi-major axis histories for Basilisk propagated state, NASA 42 propagated state, and the design reference.

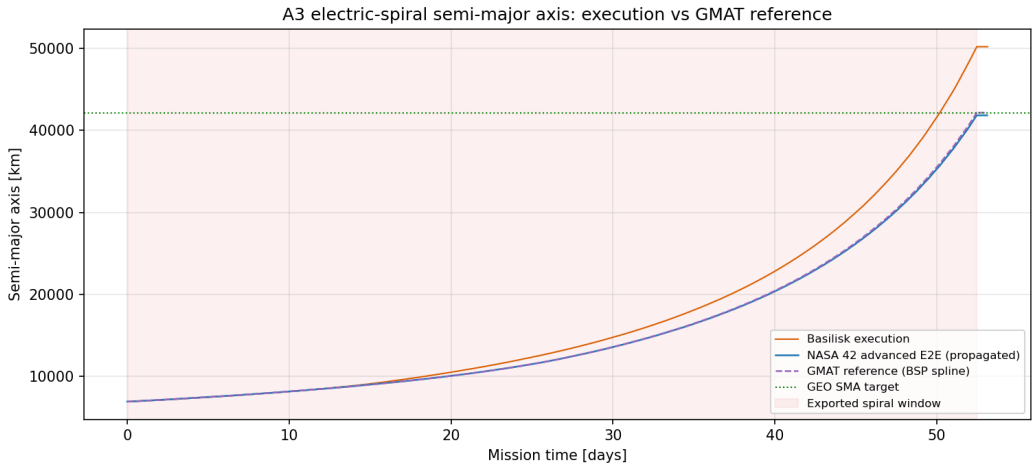


Figure 10: Semi-major axis versus mission time for Basilisk execution, NASA 42 execution, and the design reference trajectory.

Configuration item	Basilisk execution	NASA 42 execution
Burn stop logic	Exported schedule end time	Exported schedule end time
Thrust command chain	Direct body-frame thrust force	Command-scaled polynomial thrust
Eclipse / solar power	Continuous penumbra illumination factor	Binary eclipse (sun on/off)
Earth gravity harmonics	JGM2-equivalent 20×20	EGM96 18×18
Geometry / aero–SRP	Simplified bus model	Facet mesh with shadowing
Orbit propagation	Free 6-DOF with perturbations	Propagated 6-DOF with spline reference orbit
Actuators / sensors	4 RW + MTB	4 RW pyramid; 3 gyros + star tracker
Dynamics step [s]	0.5 with GNC / 10 schedule macro-step	1.0

Table 9: Execution configuration (Basilisk vs NASA 42).

1.11 Long-duration end-to-end transfer campaign (∼53 days)

The electric-spiral campaign covers ∼ 52.5 days of thrusting within a ∼ 53.2-day total mission horizon. Schedule ingestion matches the exported burn window within one second in both executors. Basilisk consumes 3.969 kg propellant (+5.4% above design); NASA 42 consumes 3.767 kg (+0.03%). Mean spiral thrust and electrical power are 17.6 mN / 0.819 kW in Basilisk and 15.0 mN / 0.713 kW in NASA 42. Basilisk closes at 50 227 km semi-major axis; NASA 42 closes at 41 844 km (Figure 10). Thrust-axis pointing peaks are 2.311° (Basilisk) and 8.72° (NASA 42).

Campaign metric	Basilisk execution	NASA 42 execution
Mission horizon [days]	53.2	53.2
Final SMA [km]	50227.1	41844.5
SMA error vs design [km]	+8063.2	−319.4
Fuel used [kg]	3.969	3.767
Burn timing error [s]	0.0	0.0
Mean spiral thrust [mN]	17.6	15.0
Mean spiral power [kW]	0.819	0.713
Thrust duty fraction [%]	100.0	87.1
Eclipse fraction [%]	0.04	12.77
Max thrust-pointing error [deg]	2.311	8.724
Thrust/eclipse pattern	Near-continuous; penumbra dips	Binary eclipse; thrust/power pulses

Table 10: End-to-end transfer campaign results (Basilisk vs NASA 42).

1.12 Constellation divergence and replay workflows

Constellation implementation across GMAT, Basilisk, and FreeFlyer establishes a common multi-spacecraft workflow pattern for divergence and replay analysis. The three implementations differ in execution architecture, but they preserve the same benchmarking intent: synchronized case progression, fixed initial-condition control per case, and archived state histories suitable for side-by-side perturbation-impact inspection.

Constellation implementation item	Value
Tools with implemented workflow	GMAT, Basilisk, FreeFlyer
GMAT constellation	5 spacecraft, synchronized propagation, per-case force models
Basilisk constellation run	5 spacecraft, mission type constellation, 72 deg phase spacing
Basilisk replay evidence	Recorded visualization session for constellation scenario
FreeFlyer mission sequence	5-case ladder, common IC restore before each case
FreeFlyer default duration [h]	24
FreeFlyer dynamics step [s]	10
FreeFlyer state export cadence [s]	600
Common reference epoch	02 Jun 2026 00:00:00 UTC

Table 11: Constellation and replay workflow implementation status across GMAT, Basilisk, and FreeFlyer.

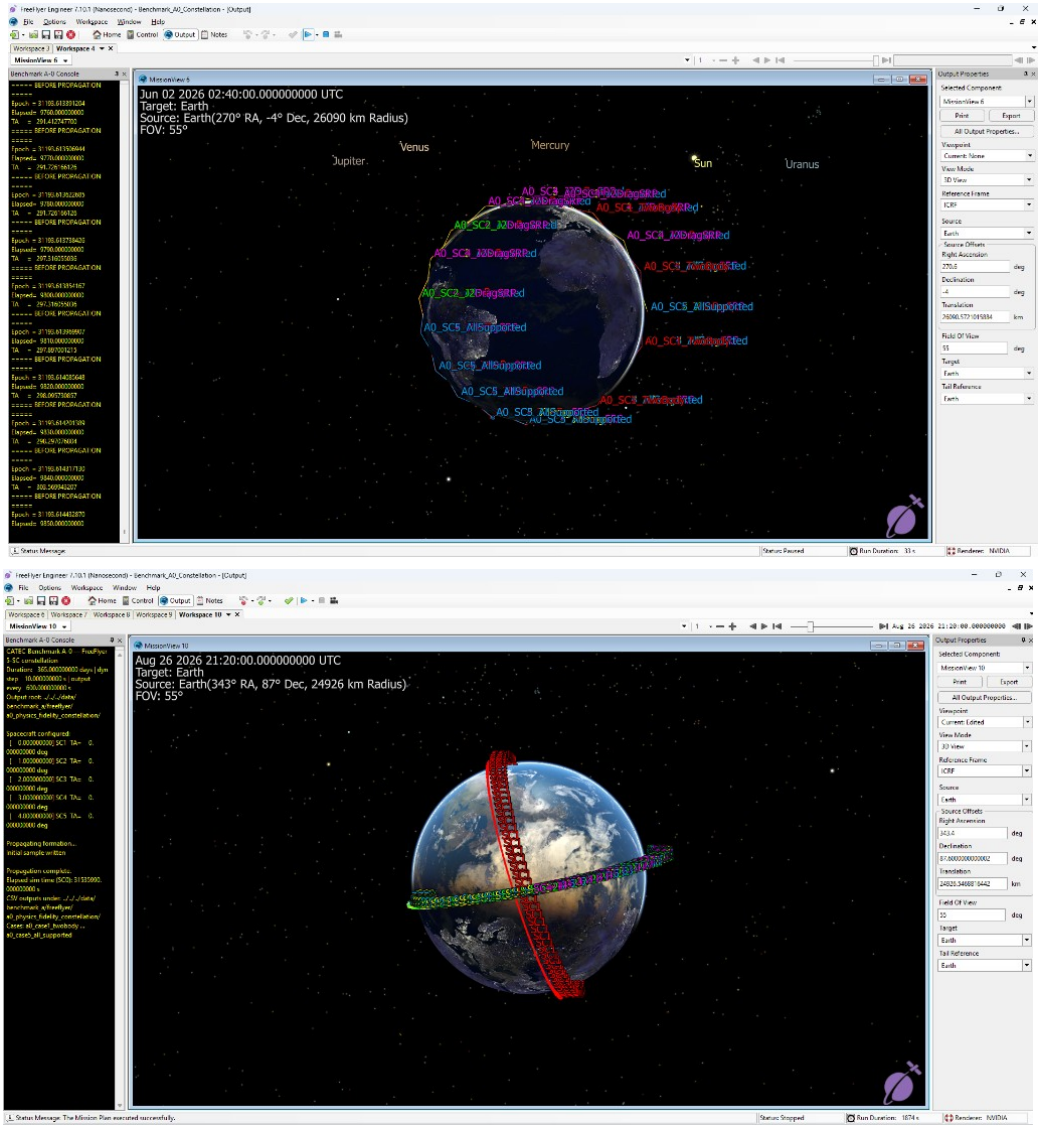


Figure 11: FreeFlyer constellation simulation for the five-spacecraft benchmark (top: early mission view with per-case force-model labels; bottom: end-of-campaign view after full propagation).

The implemented constellation workflows provide a controlled baseline for cross-case divergence analysis, where observed differences are attributable to staged force-model increments rather than mismatched initial conditions.